United Institute Of Technology
Naini,Prayagraj

A

Project Report

on

Banking Management System

Under the guidance of

Mr. Dinesh Kumar Kushwaha

(Assistant professor CSE U.I.T)

Submitted by:

University roll no	Name
2302840100205	Shanu Shrivastava
2302840100215	Shiwansh Tiwari
2302840100186	Sakshi Shukla
2302840100187	Saloni Baranwal

Table of Contents

Sr.No	Title	Page Number
1.	Abstract	1
2.	Introduction	1
3.	Objective of the project	2
4.	Modules	2-3
5.	Benifits	3
6.	Hardware and Software requirements	4
7.	Data flow diagram	5
8.	Conclusion	6
9.	Future Scope	6
10.	Bibliography	7
11.	User- Interface	7-8

1. Abstract

This project report presents the development and implementation of a **Banking System GUI** using Java, leveraging the Abstract Window Toolkit (AWT) for its graphical interface. The project aims to simulate a real-world banking system, allowing users to create accounts, deposit or withdraw money, check balances, and calculate interest. The system focuses on creating a user-friendly graphical interface to enhance user interaction with the banking operations. Key functionalities include account creation, secure transactions, and interest calculation, making it suitable for basic banking operations. This project combines both object-oriented programming concepts and graphical user interface design to offer an interactive banking experience.

---

2. Introduction

The **Banking System GUI** project is designed to offer a basic banking solution through an interactive graphical user interface. Modern banking systems require a seamless interface where users can manage their financial accounts with ease. This project is built using Java AWT, providing a simple and efficient platform for users to perform operations such as creating new accounts, depositing and withdrawing funds, checking their account balance, and adding interest for savings accounts. This project serves as a learning module for understanding GUI development in Java and integrating real-world banking operations in a simplified manner.

The purpose of this project is to mimic core banking operations in a desktop application that features buttons for operations and a display area to show information. It allows users to interact with the system securely using PIN verification, making it more realistic.

---

3. Objective of the Project

The primary objectives of this project are:

- **User-friendly Interface:** To provide a simple, user-friendly graphical interface where customers can interact with their bank accounts.
- **Basic Banking Operations:** To enable basic banking functionalities like account creation, balance checking, money deposit, withdrawal, and interest calculation for savings accounts.
- **Data Security:** To implement a secure way of verifying the user's PIN before any transaction can be performed.
- **Learning Java GUI:** To enhance understanding of Java GUI development using AWT, focusing on button interactions, layout management, and text display areas.
- **Simulating Real-World Banking:** To simulate core banking functions that are commonly used in a real-world banking environment.

---

4. Modules

The project consists of the following modules, each designed to handle a specific banking operation:

1. **Create Account Module:**
    - This module collects user data, including name, father's name, account type (Savings/Current), phone number, and an initial deposit. It also allows the user to set a PIN for security.
2. **Account Details Module:**
    - This module displays the user's account details, including balance, account type, and interest rate for savings accounts.
3. **Withdraw Money Module:**
    - This module allows the user to withdraw money from their account. The system checks for sufficient funds before processing the transaction.
4. **Deposit Money Module:**

    - This module enables the user to deposit money into their account and updates the balance accordingly.
5. **Balance Check Module:**
    - This module provides real-time account balance information to the user.
6. **Add Interest Module:**
    - This module calculates interest on savings accounts based on a user-defined number of years and adds the interest to the account balance.
7. **Exit Module:**
    - This module closes the application.

Each module is activated through a corresponding button in the graphical interface, making the system interactive and easy to navigate.

---

5. Benefits

The **Banking System GUI** offers several benefits:

- **User-Focused Design:** The intuitive graphical user interface simplifies banking operations for users.
- **Data Security:** PIN-based security ensures that only authorized users can access and modify account data.
- **Efficiency:** The project efficiently handles essential banking operations, simulating a real-world banking environment.
- **Modular Design:** The system is modular, meaning different functionalities (e.g., account creation, balance checking) are separated, making the program easy to maintain and expand.
- **Interest Calculation:** The ability to calculate interest for savings accounts introduces a dynamic feature, adding financial benefits for users.
- **Learning Tool:** This project serves as a practical educational tool for understanding object-oriented programming and graphical user interface development in Java.

---

6. Hardware and Software Requirements

To implement and run the **Banking System GUI** project, the following hardware and software requirements are necessary:

Hardware Requirements:

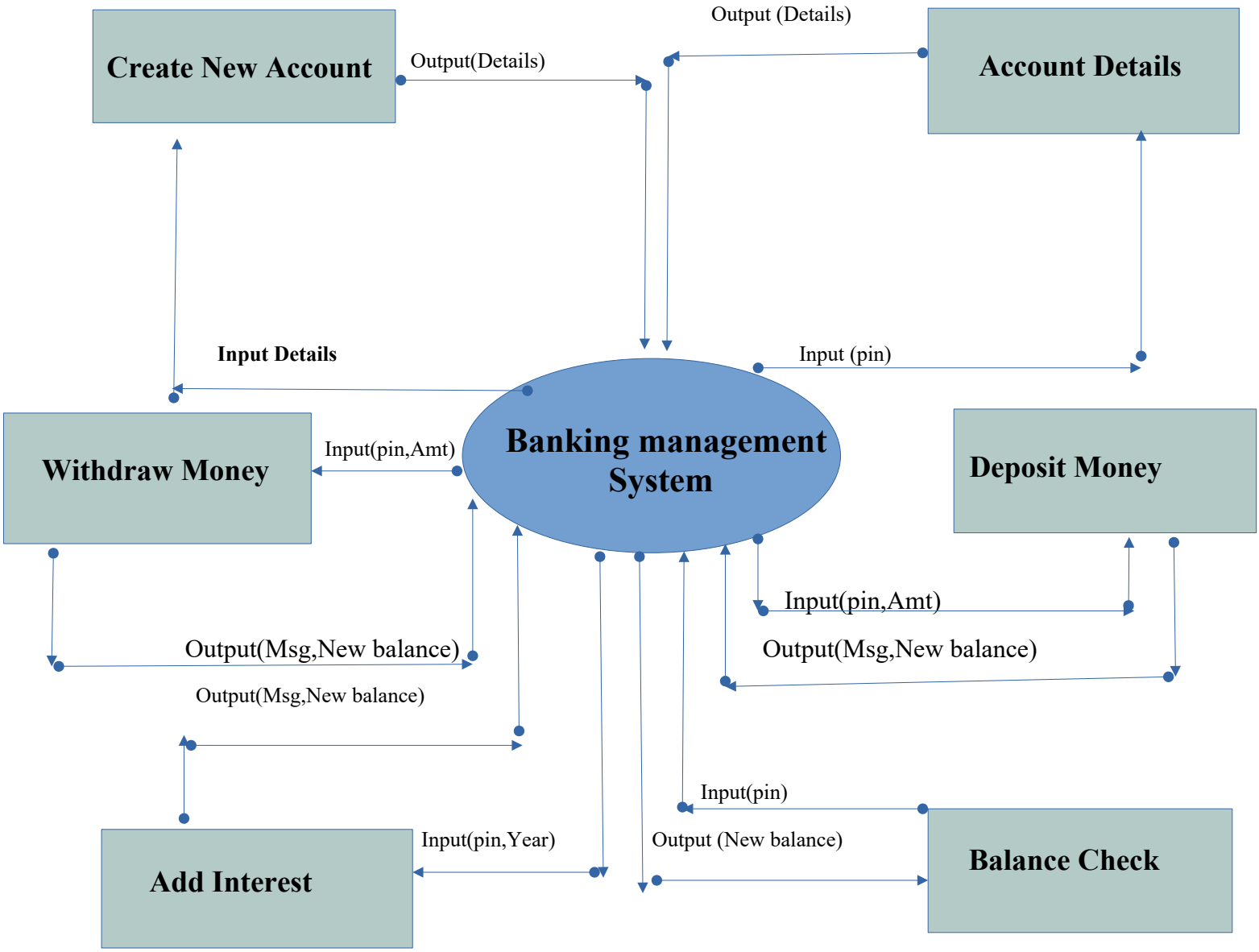
- **Processor:** Intel i3 or above
- **RAM:** 4 GB or more
- **Hard Disk:** At least 100 MB of free space
- **Display:** 1280x720 resolution or higher
- **Input Devices:** Keyboard and mouse

Software Requirements:

- **Operating System:** Windows 7/8/10, macOS, or Linux
- **Programming Language:** Java (JDK 8 or above)
- **Development Environment:**
  - **Microsoft Visual Studio Code** (or any Java-compatible IDE such as IntelliJ IDEA or Eclipse)
- **Additional Libraries:** Java Abstract Window Toolkit (AWT) for GUI development

---

7. Data Flow Diagram (DFD)



8. Conclusion

The Banking System GUI project successfully demonstrates the implementation of a simple banking application using Java's AWT framework. It provides essential functionalities such as account creation, balance checking, deposits, withdrawals, and interest calculations for savings accounts. The user interface is straightforward, allowing for easy navigation and interaction. Overall, this project serves as a foundational model for understanding basic banking operations and GUI design principles.

9. Future Scope

The project can be expanded in several ways to enhance functionality and user experience:

Database Integration: Implement a database to store account information, enabling persistent data storage and retrieval.

Enhanced Security Features: Introduce features such as password protection, encryption of sensitive data, and two-factor authentication to improve security.

User Interface Improvements: Transition to a more modern GUI framework (e.g., JavaFX) to create a visually appealing and responsive interface.

Mobile Compatibility: Develop a mobile version of the application to cater to users on various devices.

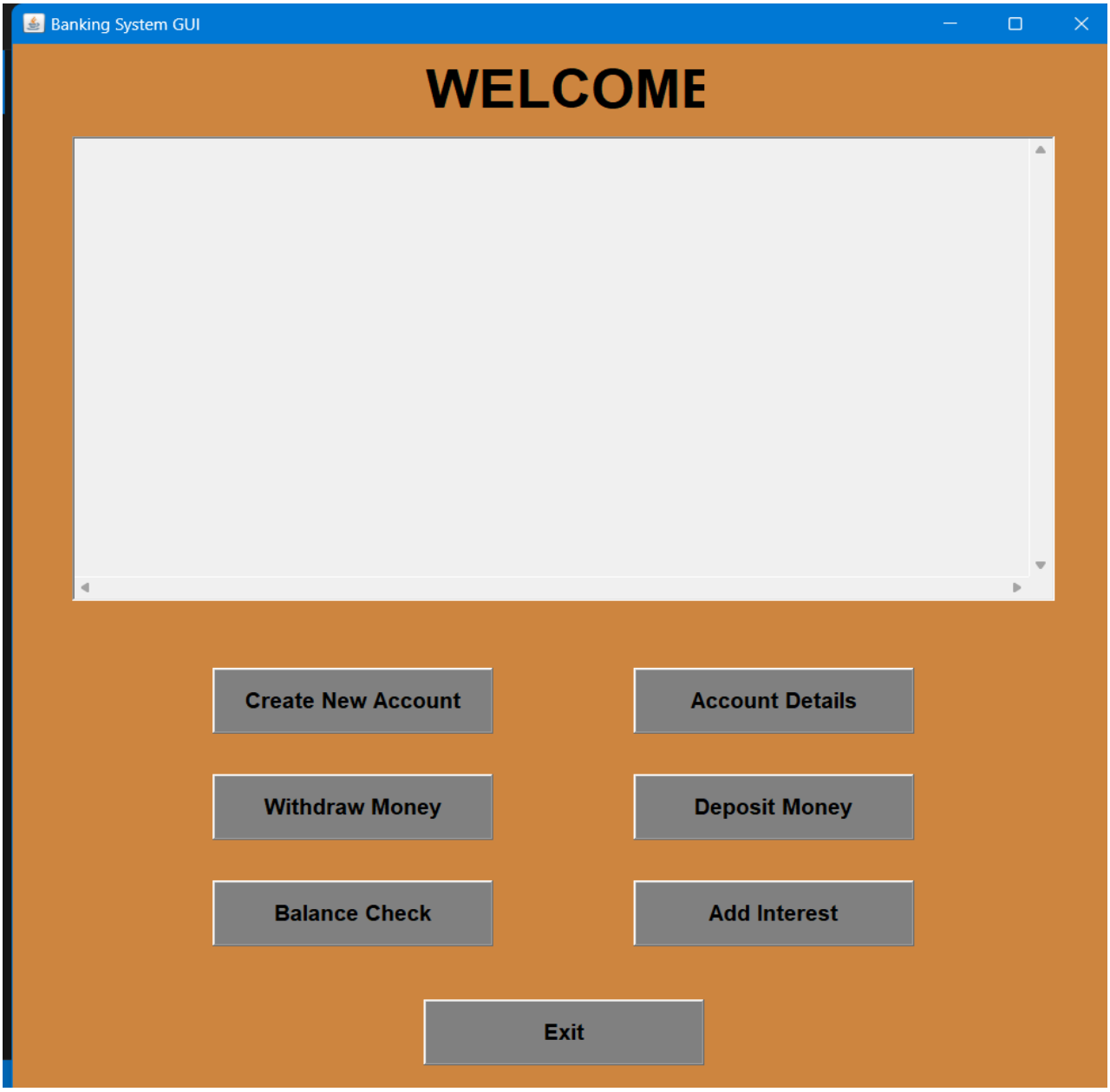
Additional Financial Services: Incorporate more banking services such as loan management, transaction history, and investment options.

Multi-user Support: Enable multiple users to access their accounts within the same application, ensuring proper session management.

10. Bibliography

- Java Development Documentation: https://docs.oracle.com/javase/8/docs/
- CodeWithHarry YouTube Channel for Java Programming: https://www.youtube.com/user/CodeWithHarry
- Apna College YouTube Channel for Java tutorials: https://www.youtube.com/c/ApnaCollege

11. User-Interface



(Before Account Creation)

(After Account Creation)